Sample Notes: Elementary Theorem Environments

November 2, 2025

1 Integers and Parity

In these notes we demonstrate the environments using elementary concepts. We refer to Definition 1.1, Lemma 1.1, Theorem 1.3, Proposition 1.4, Corollary 1.5, Claim 1.2, and Exercise 1.1.

Definition 1.1

An integer n is *even* if there exists $k \in \mathbb{Z}$ with $n = 2k$. An integer is *odd* if there exists $k \in \mathbb{Z}$ with $n = 2k + 1$.

Example 1.1.1

The integers 4, -6, and 0 are even; 3 and 7 are odd.

Lemma 1.1

If n is even, then n^2 is even.

Proof

Write $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 = (2k)^2 = 4k^2 = 2\,(2k^2)$, which is even.

Claim 1.2

The sum of two odd integers is even.

Proof

Let $a = 2r + 1$ and $b = 2s + 1$ with $r, s \in \mathbb{Z}$. Then $a + b = 2(r + s + 1)$ is even.

Theorem 1.3

For integers a and b:

- even + even = even,
- even + odd = odd,
- odd + odd = even.

Proof

Write a and b as $2r$ or $2r+1$ and add; each case follows immediately. The final case is exactly Claim 1.2.

Proposition 1.4

If $a \mid b$ and $b \mid c$ for integers a, b, c, then $a \mid c$.

Proof

There exist $m, n \in \mathbb{Z}$ with $b = am$ and $c = bn$. Then $c = (am)n = a(mn)$, so $a \mid c$.

Corollary 1.5

If n is even, then n^2 is divisible by 4.

Proof

By Lemma 1.1 write $n = 2k$. Then $n^2 = 4k^2$, a multiple of 4.

Remark. From Definition 1.1, every integer is exactly one of even or odd.

Exercise 1.1

Show that the product of two odd integers is odd.

Solution

Let $a = 2r+1$ and $b = 2s+1$. Then

$$ab = (2r+1)(2s+1) = 4rs + 2r + 2s + 1 = 2\big(2rs + r + s\big) + 1,$$

which is odd.